3.3 Graduate algebra: solutions

Read a solution only after a serious attempt. After reading, close the file and write the proof out yourself.

Warm-up

1.

The map $\beta : \mathbb{Z}/m \times \mathbb{Z}/n \to \mathbb{Z}/d$, $(\bar{a}, \bar{b}) \mapsto \overline{ab}$, is well defined (if $a' = a + km$ then $a'b - ab = kmb \in d\mathbb{Z}$ since $d \mid m$; similarly in b) and $\mathbb{Z}$-bilinear. It induces a homomorphism $\tilde{\beta} : \mathbb{Z}/m \otimes \mathbb{Z}/n \to \mathbb{Z}/d$ with $\tilde{\beta}(1 \otimes 1) = 1$, so $\tilde{\beta}$ is surjective.

On the other hand, $\bar{a} \otimes \bar{b} = ab(1 \otimes 1)$, so $T = \mathbb{Z}/m \otimes \mathbb{Z}/n$ is cyclic, generated by $1 \otimes 1$. Also $m(1 \otimes 1) = \bar{m} \otimes 1 = 0$ and $n(1 \otimes 1) = 1 \otimes \bar{n} = 0$. Write $d = um + vn$ (Bézout); then $d(1 \otimes 1) = 0$. So $|T| \leq d$. A surjection from a group of order at most d onto $\mathbb{Z}/d$ is an isomorphism.

2.

$\mathbb{Q} \otimes \mathbb{Q}/\mathbb{Z} = 0$. It suffices to kill pure tensors. Let $q \in \mathbb{Q}$ and $x \in \mathbb{Q}/\mathbb{Z}$ with $nx = 0$, $n \geq 1$. Then

$$q \otimes x = \left(n \cdot \tfrac{q}{n}\right) \otimes x = \tfrac{q}{n} \otimes nx = \tfrac{q}{n} \otimes 0 = 0.$$

$\mathbb{Q} \otimes \mathbb{Q} \cong \mathbb{Q}$. Multiplication $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ is bilinear, giving $\mu : \mathbb{Q} \otimes \mathbb{Q} \to \mathbb{Q}$ with $\mu(q \otimes r) = qr$; it is surjective ($\mu(q \otimes 1) = q$). For injectivity, take any element $\sum_i q_i \otimes r_i$ and write all $q_i = a_i/N$, $r_i = b_i/N$ over a common denominator N. Then $q_i \otimes r_i = a_i b_i \left(\frac{1}{N} \otimes \frac{1}{N}\right)$, and

$$\tfrac{1}{N} \otimes \tfrac{1}{N} = \left(N \cdot \tfrac{1}{N^2}\right) \otimes \tfrac{1}{N} = \tfrac{1}{N^2} \otimes 1.$$

So every element has the form $c\left(\frac{1}{N^2} \otimes 1\right) = \frac{c}{N^2} \otimes 1$, i.e. every element is $q \otimes 1$ for some q. Since $\mu(q \otimes 1) = q$, μ is injective.

3.

(a) A homomorphism $\varphi : \mathbb{Z}/m \to \mathbb{Z}/n$ is determined by $y = \varphi(\bar{1})$, which must satisfy $my = 0$ in $\mathbb{Z}/n$; conversely each such y defines a homomorphism. So $\mathrm{Hom}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \{y \in \mathbb{Z}/n : my = 0\}$. Now $my \equiv 0 \pmod{n}$ iff $\frac{n}{d} \mid \frac{m}{d} y$ iff $\frac{n}{d} \mid y$ (as $\gcd(\frac{m}{d}, \frac{n}{d}) = 1$). So the solutions are the multiples of n/d in $\mathbb{Z}/n$, a cyclic subgroup of order d.

(b) Let $\varphi : \mathbb{Q} \to \mathbb{Z}$ and $q \in \mathbb{Q}$. For every $n \geq 1$, $\varphi(q) = n\varphi(q/n)$ is divisible by n. The only integer divisible by every n is 0.

4.

(a) Let $\iota : 2\mathbb{Z} \to \mathbb{Z}$ be the inclusion and $\psi : 2\mathbb{Z} \to \mathbb{Z}$, $\psi(2k) = k$. If $\mathbb{Z}$ were injective, ψ would extend to $\Psi : \mathbb{Z} \to \mathbb{Z}$ with $2\Psi(1) = \Psi(2) = \psi(2) = 1$, impossible in $\mathbb{Z}$.

(b) Baer's criterion: I is injective iff every homomorphism from an ideal $\mathfrak{a} \subseteq A$ to I extends to A. The ideals of $\mathbb{Z}$ are $n\mathbb{Z}$. If $n = 0$ there is nothing to do. If $n \neq 0$ and $\psi : n\mathbb{Z} \to \mathbb{Q}$ has $\psi(n) = q$, define $\Psi : \mathbb{Z} \to \mathbb{Q}$ by $\Psi(1) = q/n$. Then $\Psi(nk) = kq = \psi(nk)$. So $\mathbb{Q}$ is injective. (The same argument shows every divisible group is injective.)

5.

S_3 has three conjugacy classes: $\{1\}$, the three transpositions, the two 3-cycles. So there are three irreducible representations, with $\sum d_i^2 = 6$, forcing degrees $1, 1, 2$. The one-dimensional ones are the trivial and sign characters. The values of the 2-dimensional character χ follow from column orthogonality with the identity column, $\sum_i \chi_i(1)\chi_i(g) = 0$ for $g \neq 1$:

- at (12): $1 - 1 + 2\chi((12)) = 0$, so $\chi((12)) = 0$;
- at (123): $1 + 1 + 2\chi((123)) = 0$, so $\chi((123)) = -1$.

	1 (1)	(12) (3)	(123) (2)
trivial	1	1	1
sign	1	−1	1
standard χ	2	0	−1

(Class sizes in parentheses.) The permutation representation on $\mathbb{C}^3$ has character $\pi = (3, 1, 0)$ (number of fixed points). Then

$$\langle \pi, \text{triv} \rangle = \tfrac{1}{6}(3 + 3 \cdot 1 + 0) = 1, \quad \langle \pi, \text{sign} \rangle = \tfrac{1}{6}(3 - 3 + 0) = 0, \quad \langle \pi, \chi \rangle = \tfrac{1}{6}(6 + 0 + 0) = 1.$$

So $\mathbb{C}^3 \cong \text{triv} \oplus \text{standard}$: the line spanned by $(1, 1, 1)$ and the plane $x_1 + x_2 + x_3 = 0$.

6.

(a) By the universal property of P applied to p_1', p_2', there is a unique $u : P' \to P$ with $p_i u = p_i'$. Symmetrically there is a unique $v : P \to P'$ with $p_i' v = p_i$. Then $p_i(uv) = p_i$ for $i = 1, 2$. The identity id_P also satisfies $p_i \,\mathrm{id}_P = p_i$, so by the uniqueness part of the universal property of P (applied to the pair p_1, p_2 themselves), $uv = \mathrm{id}_P$. Similarly $vu = \mathrm{id}_{P'}$.

(b) $A \otimes_{\mathbb{Z}} B$ is a commutative ring with $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$, and $i_A(a) = a \otimes 1$, $i_B(b) = 1 \otimes b$ are ring homomorphisms. Let $\varphi : A \to C$ and $\psi : B \to C$ be ring homomorphisms into a commutative ring C. The map $A \times B \to C$, $(a, b) \mapsto \varphi(a)\psi(b)$, is $\mathbb{Z}$-bilinear, so there is an additive $\theta : A \otimes B \to C$ with $\theta(a \otimes b) = \varphi(a)\psi(b)$. It is multiplicative on pure tensors,

$$\theta(aa' \otimes bb') = \varphi(a)\varphi(a')\psi(b)\psi(b') = \theta(a \otimes b)\,\theta(a' \otimes b')$$

(using commutativity of C), hence on all elements by distributivity, and $\theta(1 \otimes 1) = 1$. Also $\theta i_A = \varphi$ and $\theta i_B = \psi$. Uniqueness: any ring map θ' with these properties satisfies $\theta'(a \otimes b) = \theta'((a \otimes 1)(1 \otimes b)) = \varphi(a)\psi(b)$, and pure tensors generate $A \otimes B$ additively.

Core

7.

(a) ⇒ (c). Define $\eta : M \to M' \oplus M''$, $\eta(m) = (r(m), g(m))$. Injective: if $r(m) = 0$ and $g(m) = 0$, then $m = f(x)$ for some x (exactness), and $x = rf(x) = r(m) = 0$, so $m = 0$. Surjective: given (x, y), pick m_0 with $g(m_0) = y$ and put $m = m_0 + f(x - r(m_0))$. Then $g(m) = y$ (as $gf = 0$) and $r(m) = r(m_0) + x - r(m_0) = x$. Let $\theta = \eta^{-1}$. Then $\eta(f(x)) = (x, 0)$, so $\theta(x, 0) = f(x)$; and $g\theta(x, y) = y$ since the second coordinate of η is g.

(b) ⇒ (c). Define $\theta(x, y) = f(x) + s(y)$. Then $g\theta(x, y) = gs(y) = y$ and $\theta(x, 0) = f(x)$. Injective: if $f(x) + s(y) = 0$, applying g gives $y = 0$, then $f(x) = 0$, so $x = 0$. Surjective: for $m \in M$, $g(m - sg(m)) = 0$, so $m - sg(m) = f(x)$ for some x, and $m = \theta(x, g(m))$.

(c) ⇒ (a) and (b). Put $r = \pi_1\theta^{-1}$ and $s(y) = \theta(0, y)$. Then $rf(x) = \pi_1\theta^{-1}\theta(x, 0) = x$ and $gs(y) = g\theta(0, y) = y$.

Non-split example. $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$. A splitting would give $\mathbb{Z} \cong \mathbb{Z} \oplus \mathbb{Z}/2$, but $\mathbb{Z}$ has no element of order 2.

8.

Let $0 \to N' \xrightarrow{f} N \xrightarrow{g} N''$ be exact and write $f_*(\varphi) = f \circ \varphi$, $g_*(\psi) = g \circ \psi$.

- f_* *injective:* if $f\varphi = 0$ then $\varphi = 0$ since f is injective.
- $\operatorname{im} f_* \subseteq \ker g_*$*:* $gf\varphi = 0$ since $gf = 0$.
- $\ker g_* \subseteq \operatorname{im} f_*$*:* if $g\psi = 0$ then $\psi(M) \subseteq \ker g = f(N')$. Since f is an isomorphism onto its image, $\varphi = f^{-1} \circ \psi : M \to N'$ is a well-defined homomorphism with $f\varphi = \psi$.

Not exact: $\mathbb{Z} \to \mathbb{Z}/2 \to 0$ is onto, but applying $\operatorname{Hom}(\mathbb{Z}/2, -)$ gives $0 = \operatorname{Hom}(\mathbb{Z}/2, \mathbb{Z}) \to \operatorname{Hom}(\mathbb{Z}/2, \mathbb{Z}/2) \cong \mathbb{Z}/2$, which is not onto.

9.

Lemma. $X' \xrightarrow{u} X \xrightarrow{v} X'' \to 0$ is exact iff for every module P, $0 \to \operatorname{Hom}(X'', P) \xrightarrow{v^*} \operatorname{Hom}(X, P) \xrightarrow{u^*} \operatorname{Hom}(X', P)$ is exact. *Proof of "⇐":* take $P = X''/\operatorname{im} v$ and the projection π: $v^*\pi = \pi v = 0$, so $\pi = 0$ by injectivity of v^*, i.e. v is onto. Take $P = X''$: $u^*v^*(\mathrm{id}) = vu = 0$, so $\operatorname{im} u \subseteq \ker v$. Take $P = X/\operatorname{im} u$ and the projection q: $u^*q = 0$, so $q = v^*h = hv$ for some h, and then $\ker v \subseteq \ker q = \operatorname{im} u$. ("⇒" is the left exactness of $\operatorname{Hom}(-, P)$, proved like Problem 8.)

Now let $N' \to N \to N'' \to 0$ be exact. For any P, the sequence $0 \to \operatorname{Hom}(N'', \operatorname{Hom}(M, P)) \to \operatorname{Hom}(N, \operatorname{Hom}(M, P)) \to \operatorname{Hom}(N', \operatorname{Hom}(M, P))$ is exact (left exactness of $\operatorname{Hom}(-, Q)$). By the natural adjunction $\operatorname{Hom}(N, \operatorname{Hom}(M, P)) \cong \operatorname{Hom}(M \otimes N, P)$, this is the sequence $0 \to \operatorname{Hom}(M \otimes N'', P) \to \operatorname{Hom}(M \otimes N, P) \to \operatorname{Hom}(M \otimes N', P)$. (Naturality of the adjunction in N is exactly what makes the maps match.) By the lemma, $M \otimes N' \to M \otimes N \to M \otimes N'' \to 0$ is exact.

Not left exact: $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z}$ is exact, but after $\mathbb{Z}/2 \otimes_{\mathbb{Z}} -$ it becomes $\mathbb{Z}/2 \xrightarrow{2=0} \mathbb{Z}/2$, which is not injective.

10.

(a) ⇒. Let F be the free module on the set P and $\pi : F \to P$ the obvious surjection. Since P is projective, id_P lifts to $s : P \to F$ with $\pi s = \mathrm{id}_P$. By the splitting lemma, $F \cong P \oplus \ker \pi$.

⇐. A free module F with basis (e_i) is projective: given a surjection $M \to M''$ and $\varphi : F \to M''$, choose preimages m_i of $\varphi(e_i)$ and define the lift by $e_i \mapsto m_i$. If $F = P \oplus Q$ and $\varphi : P \to M''$, extend by 0 on Q, lift to $\tilde{\varphi} : F \to M$, and restrict to P.

(b) By the Chinese remainder theorem $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$ as rings, so as a module over itself $\mathbb{Z}/6 = 3\mathbb{Z}/6 \oplus 2\mathbb{Z}/6$, where $3\mathbb{Z}/6 \cong \mathbb{Z}/2$ as $\mathbb{Z}/6$-modules. So $\mathbb{Z}/2$ is a direct summand of the free module $\mathbb{Z}/6$, hence projective. It is not free: a free $\mathbb{Z}/6$-module of rank r has 6^r elements, and 2 is not a power of 6.

11.

Recall $S^{-1}f(m/s) = f(m)/s$. Since $gf = 0$, $(S^{-1}g)(S^{-1}f) = 0$. Conversely let $m/s \in S^{-1}M$ with $g(m)/s = 0$ in $S^{-1}M''$. Then $t\,g(m) = 0$ for some $t \in S$, i.e. $g(tm) = 0$, so $tm = f(m')$ for

some $m' \in M'$. Hence

$$\frac{m}{s} = \frac{tm}{ts} = \frac{f(m')}{ts} = (S^{-1}f)\left(\frac{m'}{ts}\right).$$

So the localised sequence is exact.

Flatness: if $N' \to N$ is injective, then $0 \to N' \to N$ is exact, so $0 \to S^{-1}N' \to S^{-1}N$ is exact, i.e. $S^{-1}A \otimes N' \to S^{-1}A \otimes N$ is injective (via the natural isomorphisms $S^{-1}A \otimes_A - \cong S^{-1}(-)$).

12.

(a) Suppose $M \neq 0$ and let $x_1, \dots, x_n$ ($n \geq 1$) be a generating set of minimal size. Since $x_n \in M = \mathfrak{m}M$, write $x_n = \sum_{i=1}^{n} a_i x_i$ with $a_i \in \mathfrak{m}$. Then $(1 - a_n)x_n = \sum_{i<n} a_i x_i$. As $a_n \in \mathfrak{m}$, $1 - a_n \notin \mathfrak{m}$, so it is a unit in the local ring A. Hence $x_n \in (x_1, \dots, x_{n-1})$, contradicting minimality. (If $n = 1$, this says $x_1 = 0$, so $M = 0$.)

(b) "Only if" is clear. Conversely, let $N = \sum Ax_i$ and suppose the images span $M/\mathfrak{m}M$; then $N + \mathfrak{m}M = M$. The module M/N is finitely generated and $\mathfrak{m}(M/N) = (\mathfrak{m}M + N)/N = M/N$. By (a), $M/N = 0$, i.e. $N = M$.

13.

Nilradical ⊆ every prime: if $a^n = 0 \in \mathfrak{p}$ then $a \in \mathfrak{p}$ since $\mathfrak{p}$ is prime.

Intersection ⊆ nilradical: suppose f is not nilpotent. Let Σ be the set of ideals I with $f^n \notin I$ for all $n \geq 0$. $\Sigma \ni (0)$ is non-empty, and the union of a chain in Σ is in Σ, so by Zorn Σ has a maximal element $\mathfrak{p}$. It is prime: if $x, y \notin \mathfrak{p}$, then $\mathfrak{p} + (x)$ and $\mathfrak{p} + (y)$ strictly contain $\mathfrak{p}$, so $f^m \in \mathfrak{p} + (x)$ and $f^n \in \mathfrak{p} + (y)$ for some m, n. Then $f^{m+n} \in \mathfrak{p} + (xy)$, so $\mathfrak{p} + (xy) \notin \Sigma$, so $xy \notin \mathfrak{p}$. Also $1 = f^0 \notin \mathfrak{p}$, so $\mathfrak{p}$ is proper. Since $f \notin \mathfrak{p}$, f is not in the intersection of all primes.

14.

(a) The prime ideals of the PID $\mathbb{Z}$ are (0) and (p) for primes p. Closed sets are $V(n)$ for $n \geq 0$ (every ideal is (n)). $V(0)$ is everything; $V(1) = \emptyset$; for $n \geq 2$, $V(n) = \{(p) : p \mid n\}$, a finite set of non-zero primes, and every finite set of non-zero primes arises (take n = their product). So the closed sets are $\emptyset$, the whole space, and the finite sets of non-zero primes. The closed points are the (p) (since $V(p) = \{(p)\}$). The closure of $\{(0)\}$ is $V(0)$, the whole space: (0) is a dense "generic point" and is not closed.

(b) The sets $D(f)$ form a base, so it suffices to show that every cover $\operatorname{Spec} A = \bigcup_{i \in I} D(f_i)$ has a finite subcover. The cover condition says $\bigcap_i V(f_i) = V(\{f_i\}_i) = V(\mathfrak{a}) = \emptyset$, where $\mathfrak{a}$ is the ideal generated by the f_i. If $\mathfrak{a} \neq A$, it lies in a maximal ideal, which would be a point of $V(\mathfrak{a})$. So $\mathfrak{a} = A$, and $1 = \sum_{j=1}^{n} a_j f_{i_j}$ for finitely many indices. Then $(f_{i_1}, \dots, f_{i_n}) = A$, so $V(f_{i_1}, \dots, f_{i_n}) = \emptyset$, i.e. $D(f_{i_1}) \cup \cdots \cup D(f_{i_n}) = \operatorname{Spec} A$.

15.

Suppose $I \subseteq A[x]$ is an ideal that is not finitely generated. Choose $f_1 \in I$ of least degree, and inductively $f_{k+1} \in I \setminus (f_1, \dots, f_k)$ of least degree among elements of that set (it is non-empty because I is not finitely generated). Let $n_k = \deg f_k$ and a_k the leading coefficient of f_k. By construction $n_1 \leq n_2 \leq \cdots$ (an element of $I \setminus (f_1, \dots, f_k)$ is also in $I \setminus (f_1, \dots, f_{k-1})$).

The chain $(a_1) \subseteq (a_1, a_2) \subseteq \cdots$ in the Noetherian ring A stabilises, so for some N, $a_{N+1} = \sum_{i=1}^{N} r_i a_i$ with $r_i \in A$. Let

$$g = f_{N+1} - \sum_{i=1}^{N} r_i x^{n_{N+1}-n_i} f_i.$$

The subtracted sum lies in $(f_1, \dots, f_N)$ and has degree at most n_{N+1} with $x^{n_{N+1}}$-coefficient a_{N+1}. So $\deg g < n_{N+1}$, and $g \in I \setminus (f_1, \dots, f_N)$ (else $f_{N+1} \in (f_1, \dots, f_N)$). This contradicts the choice of f_{N+1} as an element of least degree. So every ideal of $A[x]$ is finitely generated.

16.

Let $N(a + b\sqrt{-5}) = a^2 + 5b^2 = \alpha\bar{\alpha}$. N is multiplicative, and $N(\alpha) = 1$ iff $\alpha = \pm 1$ (the units).

(a) $N(2) = 4$, $N(3) = 9$, $N(1 \pm \sqrt{-5}) = 6$. The equation $a^2 + 5b^2 = 2$ or 3 has no integer solution, so there are no elements of norm 2 or 3. If $2 = \alpha\beta$ with neither a unit, then $N(\alpha)N(\beta) = 4$ with both > 1, forcing $N(\alpha) = 2$: impossible. Similarly for 3 ($9 = 3 \cdot 3$) and $1 \pm \sqrt{-5}$ ($6 = 2 \cdot 3$). So all four are irreducible. They are not associates of each other (2 and 3 have different norms from each other and from $1 \pm \sqrt{-5}$; and $2, 3$ are not associates of $1 \pm \sqrt{-5}$ as the units are ± 1). So $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ are two essentially different factorisations into irreducibles; R is not a UFD.

(b) Write $\omega = \sqrt{-5}$. Note $R \cong \mathbb{Z}[x]/(x^2 + 5)$.

Maximality. $R/\mathfrak{p} \cong \mathbb{Z}[x]/(x^2 + 5, 2, 1 + x) \cong \mathbb{F}_2[x]/(x^2 + 1, x + 1)$. In $\mathbb{F}_2[x]$, $x^2 + 1 = (x + 1)^2$, so this is $\mathbb{F}_2[x]/(x + 1) \cong \mathbb{F}_2$, a field. $R/\mathfrak{q} \cong \mathbb{F}_3[x]/(x^2 + 2, x + 1)$; at $x = -1$, $x^2 + 2 = 3 = 0$, so $x + 1 \mid x^2 + 2$ and the quotient is $\mathbb{F}_3$. Likewise $R/\mathfrak{q}' \cong \mathbb{F}_3[x]/(x^2 + 2, x - 1) \cong \mathbb{F}_3$. All three ideals are maximal.

Products.

- $\mathfrak{p}^2 = (4,\ 2 + 2\omega,\ (1 + \omega)^2) = (4,\ 2 + 2\omega,\ -4 + 2\omega)$. All generators are divisible by 2, so $\mathfrak{p}^2 \subseteq (2)$. Conversely $(2 + 2\omega) - (-4 + 2\omega) = 6$ and $6 - 4 = 2$ lie in $\mathfrak{p}^2$. So $\mathfrak{p}^2 = (2)$.
- $\mathfrak{q}\mathfrak{q}' = (9,\ 3 - 3\omega,\ 3 + 3\omega,\ (1 + \omega)(1 - \omega)) = (9,\ 3 \pm 3\omega,\ 6)$. All divisible by 3; and $9 - 6 = 3$. So $\mathfrak{q}\mathfrak{q}' = (3)$.
- $\mathfrak{p}\mathfrak{q} = (6,\ 2 + 2\omega,\ 3 + 3\omega,\ (1 + \omega)^2)$. Each generator is divisible by $1 + \omega$: $6 = (1 + \omega)(1 - \omega)$, $2 + 2\omega = 2(1 + \omega)$, $3 + 3\omega = 3(1 + \omega)$. Conversely $(3 + 3\omega) - (2 + 2\omega) = 1 + \omega \in \mathfrak{p}\mathfrak{q}$. So $\mathfrak{p}\mathfrak{q} = (1 + \omega)$.
- Applying complex conjugation (an automorphism of R fixing $\mathfrak{p}$, since $1 - \omega = 2 - (1 + \omega) \in \mathfrak{p}$, and swapping $\mathfrak{q}, \mathfrak{q}'$): $\mathfrak{p}\mathfrak{q}' = (1 - \omega)$.

Hence $(6) = (2)(3) = \mathfrak{p}^2\mathfrak{q}\mathfrak{q}'$, and also $(6) = (1+\omega)(1-\omega) = \mathfrak{p}\mathfrak{q} \cdot \mathfrak{p}\mathfrak{q}'$: the same prime factorisation.

(c) If $\mathfrak{p} = (\alpha)$ then $\alpha \mid 2$ and $\alpha \mid 1 + \omega$, so $N(\alpha)$ divides $\gcd(4, 6) = 2$. There is no element of norm 2, so $N(\alpha) = 1$, α is a unit and $\mathfrak{p} = R$, contradicting $R/\mathfrak{p} \cong \mathbb{F}_2$.

17.

Let $K = \mathbb{Q}(\sqrt{d})$ and $\alpha = a + b\sqrt{d}$ with $a, b \in \mathbb{Q}$.

If $b = 0$: $\alpha = a \in \mathbb{Q}$ is integral over $\mathbb{Z}$ iff $a \in \mathbb{Z}$ ($\mathbb{Z}$ is integrally closed, by the rational root test).

If $b \neq 0$: the minimal polynomial of α over $\mathbb{Q}$ is $x^2 - 2ax + (a^2 - db^2)$. By the fact quoted, α is integral iff $2a \in \mathbb{Z}$ and $a^2 - db^2 \in \mathbb{Z}$. (Conversely if both hold, α satisfies a monic integer polynomial.)

Put $A = 2a \in \mathbb{Z}$. Then $4(a^2 - db^2) = A^2 - d(2b)^2 \in 4\mathbb{Z}$, so $d(2b)^2 \in \mathbb{Z}$. Write $2b = p/q$ in lowest terms; then $q^2 \mid dp^2$, so $q^2 \mid d$, and since d is squarefree, $q = 1$. So $B = 2b \in \mathbb{Z}$, and the condition is

$$A^2 \equiv dB^2 \pmod 4.$$

Squares mod 4 are 0 and 1.

- If $d \equiv 2$ or $3 \pmod 4$: if B is odd then $A^2 \equiv d \pmod 4$, impossible. So B is even, then $A^2 \equiv 0$, so A is even. Hence $a, b \in \mathbb{Z}$, and conversely all of $\mathbb{Z}[\sqrt{d}]$ is integral. The ring of integers is $\mathbb{Z}[\sqrt{d}]$.
- If $d \equiv 1 \pmod 4$: the condition is $A^2 \equiv B^2 \pmod 4$, i.e. $A \equiv B \pmod 2$. So the integral elements are $\frac{A+B\sqrt{d}}{2}$ with $A \equiv B \pmod 2$. These are exactly the elements $m + n\frac{1+\sqrt{d}}{2}$ with $m, n \in \mathbb{Z}$ (take $n = B$, $m = (A - B)/2$). So the ring of integers is $\mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$.

18.

Let $\mathfrak{m}$ be maximal and $L = \mathbb{C}[x_1, \dots, x_n]/\mathfrak{m}$, a field containing $\mathbb{C}$. As a $\mathbb{C}$-vector space it is spanned by the images of the countably many monomials, so $\dim_{\mathbb{C}} L$ is countable.

Suppose $t \in L$ is transcendental over $\mathbb{C}$. Then $\mathbb{C}[t] \subseteq L$ is a polynomial ring and $\mathbb{C}(t) \subseteq L$. The elements $\frac{1}{t-a}$, $a \in \mathbb{C}$, are linearly independent over $\mathbb{C}$: if $\sum_{j=1}^{k} c_j \frac{1}{t-a_j} = 0$ with distinct a_j, multiply by $\prod_j (t - a_j)$ to get $\sum_j c_j \prod_{l \neq j} (t - a_l) = 0$ in $\mathbb{C}[t]$; evaluating this polynomial at $t = a_j$ gives $c_j \prod_{l \neq j}(a_j - a_l) = 0$, so $c_j = 0$. This is an uncountable linearly independent set in a space of countable dimension, a contradiction.

So every element of L is algebraic over $\mathbb{C}$, and as $\mathbb{C}$ is algebraically closed, $L = \mathbb{C}$. Let $a_i \in \mathbb{C}$ be the image of x_i. Then $x_i - a_i \in \mathfrak{m}$, so $(x_1 - a_1, \dots, x_n - a_n) \subseteq \mathfrak{m}$. The left side is maximal (the quotient is $\mathbb{C}$, via evaluation at a), so equality holds.

19.

($\Rightarrow$) Let V be irreducible. For each g, $\rho(g)$ commutes with every $\rho(h)$ (as G is abelian), so it is a G-map $V \to V$, hence a scalar by Schur's lemma. Then every subspace is invariant, so irreducibility forces $\dim V = 1$.

($\Leftarrow$) If all irreducibles are one-dimensional, the regular representation $\mathbb{C}[G]$ is a direct sum of one-dimensional representations, $\rho_{\text{reg}}(g)$ is simultaneously diagonal in a suitable basis, and so all $\rho_{\text{reg}}(g)$ commute. The regular representation is faithful ($g \cdot 1 = g \neq 1$ for $g \neq 1$), so $g \mapsto \rho_{\text{reg}}(g)$ is injective and G is abelian. (Alternatively: $\sum d_i^2 = |G|$ with all $d_i = 1$ means there are $|G|$ conjugacy classes, so all classes are singletons.)

20.

$D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs^{-1} = r^{-1} \rangle$. Conjugacy classes: $\{1\}$, $\{r^2\}$, $\{r, r^3\}$, $\{s, sr^2\}$, $\{sr, sr^3\}$: five classes. The commutator subgroup is $\{1, r^2\}$ (it contains $[r, s] = r^2$, and $D_4/\langle r^2 \rangle$ has order 4, hence is abelian). So there are $[G : G'] = 4$ one-dimensional characters (Problem 25, or directly: homomorphisms to $\{\pm 1\}$ determined by the signs of r and s), and one more irreducible of degree d with $4 + d^2 = 8$, so $d = 2$. Its values follow from column orthogonality against the degree column: $\sum_i \chi_i(1)\chi_i(g) = 0$ for $g \neq 1$.

$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$. Conjugacy classes: $\{1\}$, $\{-1\}$, $\{\pm i\}$, $\{\pm j\}$, $\{\pm k\}$. Commutator subgroup $\{\pm 1\}$ ($iji^{-1}j^{-1} = -1$), with quotient the Klein four-group. Again four linear characters (determined by signs of i and j, with $\chi(k) = \chi(i)\chi(j)$) and one of degree 2.

Matching classes $1 \leftrightarrow 1$, $r^2 \leftrightarrow -1$, $r \leftrightarrow i$, $s \leftrightarrow j$, $sr \leftrightarrow k$, both tables are:

	1	r^2 / -1	r / $\pm i$	s / $\pm j$	sr / $\pm k$
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-2	0	0	0

(The linear characters are trivial on r^2, resp. -1, since it lies in the commutator subgroup. For χ_5: at r^2, $1+1+1+1+2\chi_5 = 0$ gives -2; at the other classes the four linear characters sum to 0, giving $\chi_5 = 0$.)

Not isomorphic. D_4 has five elements of order 2 (r^2, s, sr, sr^2, sr^3), while Q_8 has only one (-1).

Qualifying-exam level

21.

The ideals $\ker\varphi \subseteq \ker\varphi^2 \subseteq \ker\varphi^3 \subseteq \cdots$ form an ascending chain, so there is n with $\ker\varphi^n = \ker\varphi^{n+1}$. Let $x \in \ker\varphi$. Since φ^n is surjective, $x = \varphi^n(y)$ for some y. Then $\varphi^{n+1}(y) = \varphi(x) = 0$, so $y \in \ker\varphi^{n+1} = \ker\varphi^n$, so $x = \varphi^n(y) = 0$. Hence $\ker\varphi = 0$.

22.

(a) For $n \geq 2$ let $I_n = \{f \in R : f = 0 \text{ on } [0, 1/n]\}$, an ideal. Then $I_2 \subseteq I_3 \subseteq \cdots$, and the inclusions are strict: the continuous function $f(x) = \max\left(0, x - \frac{1}{n+1}\right)$ lies in I_{n+1} but not in I_n (it is positive at $x = \frac{1}{n}$). An ascending chain that does not stabilise means R is not Noetherian.

(b) *Each $\mathfrak{m}_x$ is maximal:* evaluation $f \mapsto f(x)$ is a surjective ring homomorphism $R \to \mathbb{R}$ with kernel $\mathfrak{m}_x$, and $\mathbb{R}$ is a field.

Every maximal ideal is some $\mathfrak{m}_x$: let $\mathfrak{m}$ be maximal and suppose $\mathfrak{m} \neq \mathfrak{m}_x$ for every x. Since $\mathfrak{m}_x$ is maximal, $\mathfrak{m} \not\subseteq \mathfrak{m}_x$, so for each x there is $f_x \in \mathfrak{m}$ with $f_x(x) \neq 0$, hence $f_x \neq 0$ on an open neighbourhood U_x of x. By compactness, $[0,1] = U_{x_1} \cup \cdots \cup U_{x_k}$. Then $g = \sum_i f_{x_i}^2 \in \mathfrak{m}$ and $g > 0$ everywhere on $[0,1]$. So $1/g \in R$ and $1 = g \cdot (1/g) \in \mathfrak{m}$, a contradiction.

23.

Method 1 (structure). $\mathbb{C} \cong \mathbb{R}[x]/(x^2+1)$. Tensor products commute with quotients: $\mathbb{R}[x]/(f) \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[x]/(f)$ as rings (by right exactness applied to $0 \to (f) \to \mathbb{R}[x] \to \mathbb{R}[x]/(f) \to 0$, together with $\mathbb{R}[x] \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[x]$). So

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[x]/(x^2+1) = \mathbb{C}[x]/((x-i)(x+i)) \cong \mathbb{C}[x]/(x-i) \times \mathbb{C}[x]/(x+i) \cong \mathbb{C} \times \mathbb{C},$$

using the Chinese remainder theorem ($(x-i)$ and $(x+i)$ are comaximal since their difference $2i$ is a unit).

Method 2 (explicit). Define $\Phi : \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \to \mathbb{C} \times \mathbb{C}$ by $\Phi(a \otimes b) = (ab, \bar{a}b)$, induced by the $\mathbb{R}$-bilinear map $(a,b) \mapsto (ab, \bar{a}b)$. It is a ring homomorphism (check on pure tensors: $\Phi(aa' \otimes bb') =$

$(aa'bb', \overline{aa'}bb') = \Phi(a \otimes b)\Phi(a' \otimes b')$, and $\Phi(1 \otimes 1) = (1, 1)$). Both sides have real dimension 4. Φ is onto: $\Phi(1 \otimes 1) = (1, 1)$ and $\Phi(i \otimes i) = (-1, 1)$ give $(1, 0)$ and $(0, 1)$ as real combinations, and $\Phi(1 \otimes i) = (i, i)$, $\Phi(i \otimes 1) = (i, -i)$ give $(i, 0), (0, i)$. A surjective linear map between real spaces of the same finite dimension is bijective.

24.

Let $W = \mathrm{Res}^G_A V$. Since A is abelian, W is a direct sum of one-dimensional representations of A (Maschke and Problem 19). Pick one, a character $\lambda : A \to \mathbb{C}^\times$ with $\mathrm{Hom}_A(\lambda, W) \neq 0$. By Frobenius reciprocity,

$$\mathrm{Hom}_G(\mathrm{Ind}^G_A \lambda, V) \cong \mathrm{Hom}_A(\lambda, \mathrm{Res}^G_A V) \neq 0.$$

So there is a non-zero G-map $\mathrm{Ind}^G_A \lambda \to V$. Its image is a non-zero subrepresentation of the irreducible V, so the map is onto. Hence $\dim V \leq \dim \mathrm{Ind}^G_A \lambda = [G : A] \cdot 1$.

25.

A one-dimensional representation is a homomorphism $\chi : G \to GL_1(\mathbb{C}) = \mathbb{C}^\times$. Since $\mathbb{C}^\times$ is abelian, χ kills every commutator, so it factors uniquely through $G^{ab} = G/G'$; conversely every homomorphism $G/G' \to \mathbb{C}^\times$ pulls back to G. Isomorphism classes of one-dimensional representations correspond to the homomorphisms themselves (two one-dimensional representations are isomorphic iff their characters are equal). So the number of one-dimensional representations of G equals the number for the abelian group G/G'. For an abelian group H, all irreducibles are one-dimensional (Problem 19) and $\sum d_i^2 = |H|$, so there are exactly $|H|$ of them. Hence the number is $|G/G'| = [G : G']$.

S_4. Conjugacy classes correspond to cycle types: 1, (12), $(12)(34)$, (123), (1234); so there are 5 irreducibles. The commutator subgroup is A_4: $S_4/A_4 \cong \mathbb{Z}/2$ is abelian, so $S_4' \subseteq A_4$; and every 3-cycle is a commutator, e.g. $(12)(13)(12)^{-1}(13)^{-1} = \big((12)(13)\big)^2 = (132)^2 = (123)$ (composing right to left), and 3-cycles generate A_4. So $[S_4 : S_4'] = 2$: there are two one-dimensional representations (trivial and sign). The other three have degrees $d_3, d_4, d_5 \geq 2$ with $d_3^2 + d_4^2 + d_5^2 = 24 - 2 = 22$. Among squares ≥ 4 that are at most 22 (namely $4, 9, 16$), the only way to write 22 as a sum of three is $4 + 9 + 9$. So the degrees are $1, 1, 2, 3, 3$.